

1. Given two vectors, x and y . The x values begin at 0 and end at 15π , with a spacing of $\frac{\pi}{100}$, while the y values begin at 0 and end at 12, with a spacing of $\frac{\pi}{10}$.
 - (a) Find size, bytes and class of x and y .
 - (b) Save input and output using 'diary'.
2. Create a vector x of values from 0 to 10π , with a spacing of $\frac{\pi}{100}$. Define vector y as

$$y = \pi \sin(x)$$

- (a) Find size, bytes and class of x and y .
 - (b) Save input and output using 'diary'.
3. Given $y1 = 2 \cos(x)$ and $y2 = 5 \sin(x)$. Let x vary from 0 to 2π in increments of 0.1π .
 - (a) Save and load workspace variables.
 - (b) Create new script, load the data saved in the workspace and plot x versus $y1$.
 - (c) Create new script, load the data saved in the workspace and plot x versus $y2$.(you may use 'doc' command for help)
 4. Given $y1 = 2x$ and $y2 = 5 \sin(x)$. Let x vary from 0 to π in increments of 0.1π .
 - (a) Save and load workspace variables.
 - (b) Create new script, load the data saved in the workspace and plot x versus $y1$.
 - (c) Create new script, load the data saved in the workspace and plot x versus $y2$.
 5. Create a vector x of values from 0 to 2π , with a spacing of $\frac{\pi}{4}$. Define vector y as

$$y = \sin(x)$$

Next, define a new vector, xq of values from 0 to 2π , with a spacing of $\frac{\pi}{16}$.

- (a) Find the yq values corresponding to the xq values by linear interpolation.
 - (b) On the same figure, plot the original y vs. x as circles, and yq vs. xq as a line.
 - (c) Repeat the exercise in part (a) and (b) using the spline(...) function to interpolate.
6. Create a vector x of values from 1 to 9. Define vector v as

$$v = [0 \ 1.41 \ 2 \ 1.41 \ 0 \ -1.41 \ -2 \ -1.41 \ 0]$$

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Next, define a new vector xq of values from 1.5 to 8.5.

- (a) Find the yq values corresponding to the xq values by linear interpolation.
- (b) On the same figure, plot the original y vs. x as circles, and yq vs. xq as a line.
- (c) Repeat the exercise in part (a) and (b) using the spline(...) function to interpolate.

1. A factorial is the product of all the integers from 1 to N . In Matlab, find the Factorial for $N = 5$.

2. Use a *for* loop to create a vector containing the first 10 elements in the harmonic series, *i.e.*,

$$\frac{1}{1} \quad \frac{1}{2} \quad \frac{1}{3} \quad \cdots \quad \frac{1}{10}$$

3. Use a *for* loop to create a vector containing the first 10 elements in the alternating harmonic series, *i.e.*,

$$\frac{1}{1} \quad -\frac{1}{2} \quad \frac{1}{3} \quad -\frac{1}{4} \cdots -\frac{1}{10}$$

4. Write Matlab code to calculate the sum of series:

$$S = 1 - \frac{x}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} \quad \text{for } x = 1.5.$$

5. Write the Matlab program to obtain the sum of all even number from 0 to 20 using *for* loop statement.

6. Write the program to find the average value of given any 10 number using *for* loop statement. Check the program with the following values : 35, 24, 5, 6, 4, 10, 23, 45, 2

7. Repeat the preceding problem, this time using a *while* loop.

8. Write a program, using *while* loop, for finding square of integers less than 5.

9. Write the Matlab program, to test whether or not π^e is greater than, or equal to, e^π .

10. In Matlab, compute the Fibonacci sequence defined by $f_1 = 0, f_2 = 1$ and $f_n = f_{n-1} + f_{n-2}$. Using *for* loop and *if* condition.

11. Draw graphs of $\sin(nx)$ on the interval $[-1, 1]$ for $n = 1, 2, 3, 4, 5, 6, 7, 8$.

12. For any n build the $n \times n$ matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \cdots & 0 \\ 1 & 1 & 0 \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ 0 & \cdots & 1 & 1 \end{bmatrix}$$

in Matlab.

Practice Problems : Eng Prac in Maths and Comput (MA 122)
B.Tech. Ist Year, 2025
Course Instructor: MKR+SK

1. Consider the following initial value problem:

$$\frac{dy}{dt} = -2ty^2, \quad y(t=0) = 1$$

Write a MATLAB code to compute $y(10)$ using Euler method with $\Delta t = 0.2$.

2. Consider the following system of equations:

$$\begin{aligned}\frac{dy}{dt} &= z \\ \frac{dz}{dt} + 5tz + (y+7)\sin t &= 0 \\ y(t=0) &= 5, \quad z(t=0) = 1.4\end{aligned}$$

Write a MATLAB code to compute the solutions of the above system using the Euler method at $t = 10$ with a time step of $\Delta t = 0.5$.

1: Consider the following data set:

x	y
1	2.1
2	2.9
3	3.7
4	4.1
5	5.0

(i): Write a Matlab code using Least Squares Linear Regression to fit the linear model $y = a_0 + a_1x$.

(ii): Calculate the coefficient of determination (r^2).

(iii): Plot both the data and the fitted curve.

2: Consider the following data set:

x	y
0	1.0
1	2.7
2	5.8
3	10.1
4	17.2

(i): Write a Matlab code using Least Squares Linear Regression to fit the quadratic model $y = a_0 + a_1x + a_2x^2$.

(ii): Compare S_y and $S_{y/x}$.

(iii): Plot both the data and the fitted curve.
